

Jiwon Kang

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Research Interests

Having started my research career in 3D vision, I expanded my focus during my Master's to Generative AI, including its practical applications and Unified Multimodal Models (UMMs). Recently, I have become interested in the challenge of world modeling by combining 3D vision capabilities with generative knowledge, which can be extended to real-world robotics and immersive gaming environments. Key research areas include:

- Diffusion & Multimodal Generative AI (Image/Video/Action/Audio Generation, Guidance, Sampling)
- Unified Multimodal Models
- 3D Vision (Novel View Synthesis, NeRF, Gaussian Splatting)

Education

M.S. in Artificial Intelligence

Kim Jaechul Graduate School of AI, KAIST

Mar. 2025 – Feb. 2027 (Expected)

Advisor: [Seungryong Kim](#) (CVLAB)

B.S. in Computer Science and Engineering

Korea University

Mar. 2019 – Feb. 2025

- GPA: 4.24 / 4.5
- Double Major in Statistics
- Leave of Absence for R.O.K. Military Service (Dec. 2020 – Jun. 2022)
- Undergraduate Research Intern @ [CVLAB](#) (Sep. 2022 – Feb. 2025), Advisor: Seungryong Kim

Experience

Kakao

Research Intern

Mar. 2026 – May 2026

Seoul, Korea

- Worked on Text-to-Speech research, repurposing Whisper as a continuous tokenizer for autoregressive diffusion-based TTS model [\[P2\]](#).

Projects

SAMSUNG

Research Collaboration

Aug. 2025 – Oct. 2025

Seoul, Korea

- Collaborated with Samsung Electronics on background face anonymization via generative face-swapping [\[C3\]](#).

Publications

*: Equal Contribution, †: Corresponding Author

International Conference

[\[C4\]](#) **Transferability Between Understanding and Generation in Unified Multimodal Models** [Paper](#) | [Project Page](#) | [Code](#)

[Jiwon Kang](#)^{*}, Heeji Yoon^{*}, Jaewoo Jung, Jaewon Min, Minkyong Jeon, Biyeon Hwang, Sangwon Jung, Seungryong Kim[†]

- Investigated how understanding and generation capabilities (e.g., counting) transfer to one another within unified multimodal models.
- Showed that transferability depends on the model architecture, and proposed a practical post-training strategy for unified multimodal models that leverages transferability.

European Conference on Computer Vision (ECCV 2026).

[C3] APPLE: Attribute-Preserving Pseudo-Labeling for Diffusion-Based Face Swapping [Paper](#) | [Project Page](#) | [Code](#)

Jiwon Kang^{*}, Yeji Choi^{*}, JoungBin Lee, Wooseok Jang, Jinhyeok Choi, Taekeun Kang, Yongjae Park, Myungin Kim, Seungryong Kim[†]

- Prior diffusion-based face swapping methods focused on changing facial identity, failing to preserve facial attributes.
- Proposed a novel pseudo-labeling framework that preserves facial attributes while swapping identities, achieving Pareto-optimal results.

IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2026).

[C2] A Noise is Worth Diffusion Guidance [Paper](#) | [Project Page](#) | [Code](#)

Donghoon Ahn^{*}, Jiwon Kang^{*}, Sanghyun Lee, Jaewon Min, Wooseok Jang, Minjae Kim, Hyungwon Cho, Sayak Paul, SeonHwa Kim, Eunju Cha[†], Kyong Hwan Jin[†], Seungryong Kim[†]

- Showed that classifier-free guidance can be replaced by refining the initial noise.
- Enabled high-quality generation without a guidance pass, highlighting the importance of the initial noise in diffusion models.

International Conference on Learning Representations (ICLR 2026).

[C1] Where and How to Perturb: On the Design of Perturbation Guidance in Diffusion and Flow Models [Paper](#) | [Project Page](#) | [Code](#)

Donghoon Ahn^{*}, Jiwon Kang^{*}, Sanghyun Lee, Minjae Kim, Wooseok Jang, Jaewon Min, Sangwu Lee, Sayak Paul, Seungryong Kim[†]

- Systematically studied perturbation guidance in diffusion and flow models.
- Showed that perturbing different layers/heads in a diffusion model exhibits different effects on the generated samples, which can be systematically combined to improve user-specified attributes via the proposed framework.

Conference on Neural Information Processing Systems (NeurIPS 2025).

Workshop

[W1] Self-Evolving Neural Radiance Fields [Paper](#) | [Project Page](#) | [Code](#)

Jaewoo Jung^{*}, Jisang Han^{*}, Jiwon Kang^{*}, Seongchan Kim, Min-Seop Kwak, Seungryong Kim[†]

- Proposed a self-training framework that lets a NeRF improve itself from few input views via pseudo-label distillation.

ICCV 2025 Workshop on Wild3D: 3D Modeling, Reconstruction, and Generation in the Wild.

Preprints & Under Review

[P2] Exploring Whisper as a Continuous Tokenizer for Text-to-Speech

Jiwon Kang, Heeji Yoon, Woo-Young Kang, Sunghun Kang, Jihwan Eom, Jaeyeol Jeon, Byungseok Roh, Seungryong Kim[†]

- Inspired by the success of RAE in the vision domain, explored whether ASR-supervised audio representations (Whisper) could be repurposed as generative representations for TTS.
- Developed (1) a Whisper-based continuous tokenizer and (2) an autoregressive diffusion-based TTS model that generates Whisper tokens, enabling high-quality speech synthesis with controllable prosody.

Under Review.

[P1] RAIN-GS: Relaxing Accurate Initialization Constraint for 3D Gaussian Splatting [Paper](#) | [Project Page](#) | [Code](#)

Jaewoo Jung^{*}, Jisang Han^{*}, Honggyu An^{*}, Jiwon Kang^{*}, Seonghoon Park^{*}, Seungryong Kim[†]

- Relaxed the accurate point-cloud initialization requirement of 3D Gaussian Splatting, enabling training from sparse or random initialization.

Under Review.

Honors

Outstanding Paper Silver Award, *IPIU 2024*

2024

- **Paper:** *Self-Evolving Neural Radiance Fields*

Extracurricular Activities

Military Service (KATUSA, Korean Augmentation to the U.S. Army)

Dec. 2020 – Jun. 2022

- MOS : 92Y (Unit Supply Specialist)
- 194th DSSB, 2nd Infantry Division, Eighth U.S. Army, Camp Humphreys

AIKU (Artificial Intelligence Society of Korea University)

Sep. 2022 – Aug. 2023

- Co-founder & Vice President of academic research society for undergraduate students